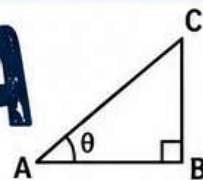


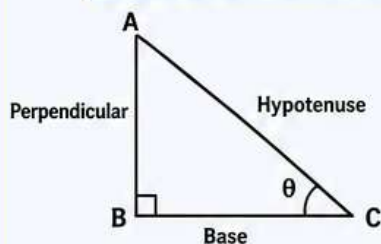


MATHS PATHSHAKA



CLASS 10 – TRIGONOMETRY FORMULA SHEET

1. BASIC RATIOS



- $\sin \theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{P}{H}$
- $\cos \theta = \frac{\text{Base}}{\text{Hypotenuse}} = \frac{B}{H}$
- $\tan \theta = \frac{\text{Perpendicular}}{\text{Base}} = \frac{P}{B}$
- $\text{cosec } \theta = \frac{\text{Hypotenuse}}{\text{Perpendicular}} = \frac{H}{P}$
- $\sec \theta = \frac{\text{Hypotenuse}}{\text{Base}} = \frac{H}{B}$
- $\cot \theta = \frac{\text{Base}}{\text{Perpendicular}} = \frac{B}{P}$

2. RECIPROCAL IDENTITIES

- $\text{cosec } \theta = \frac{1}{\sin \theta}$
- $\sec \theta = \frac{1}{\cos \theta}$
- $\cot \theta = \frac{1}{\tan \theta}$
- $\sin \theta = \frac{1}{\text{cosec } \theta}$
- $\cos \theta = \frac{1}{\sec \theta}$
- $\tan \theta = \frac{1}{\cot \theta}$

3. QUOTIENT IDENTITIES

- $\tan \theta = \frac{\sin \theta}{\cos \theta}$
- $\cot \theta = \frac{\cos \theta}{\sin \theta}$
- $\sec \theta = \frac{1}{\cos \theta}$
- $\text{cosec } \theta = \frac{1}{\sin \theta}$

4. PYTHAGOREAN IDENTITIES

- $\sin^2 \theta + \cos^2 \theta = 1$
- $1 + \tan^2 \theta = \sec^2 \theta$
- $1 + \cot^2 \theta = \text{cosec}^2 \theta$

5. TRIGONOMETRIC RATIOS OF COMPLEMENTARY ANGLES

- $\sin (90^\circ - \theta) = \cos \theta$
- $\cos (90^\circ - \theta) = \sin \theta$
- $\tan (90^\circ - \theta) = \cot \theta$
- $\cot (90^\circ - \theta) = \tan \theta$
- $\sec (90^\circ - \theta) = \text{cosec } \theta$
- $\text{cosec } (90^\circ - \theta) = \sec \theta$

6. TRIGONOMETRIC RATIOS OF SOME SPECIFIC ANGLES

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined
$\text{cosec } \theta$	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined
$\cot \theta$	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

7. IMPORTANT FORMULAS

- $(\sin \theta)^2 + (\cos \theta)^2 = 1$
- $(\cos \theta)^2 + (\sin \theta)^2 = 1$
- $\tan \theta = \frac{\sin \theta}{\cos \theta}$
- $1 - \sin^2 \theta = \cos^2 \theta$
- $1 - \cos^2 \theta = \sin^2 \theta$
- $\tan \theta \times \cot \theta = 1$
- $\sin \theta \cdot \text{cosec } \theta = 1$
- $\cos \theta \cdot \sec \theta = 1$



Remember:

H = Hypotenuse
P = Perpendicular
B = Base
 θ = Angle



JOIN NOW!

A STEP TODAY, A TOPPER TOMORROW!



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